

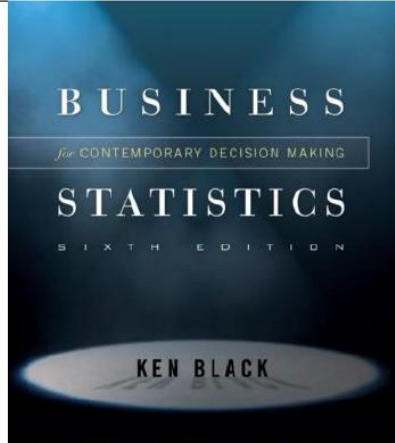
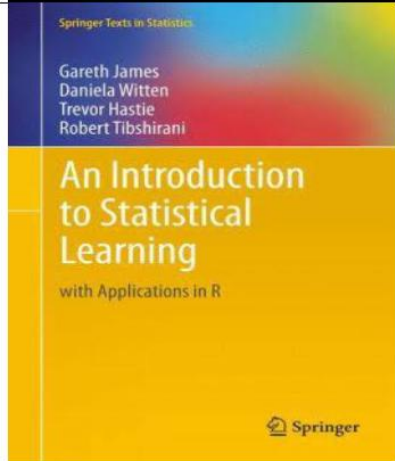
DATA SCIENCE – WHAT IS IT?



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Books and References

- Learning Data Science : <https://learningds.org/intro.html>
- Veridical Data Science : <https://vdsbook.com/>
- Python Data Science Handbook: <https://jakevdp.github.io/PythonDataScienceHandbook/>

For topics related to Statistics	Business Statistics for Contemporary Decision Making Author: Ken Black (Available online – 6 th Edition)	
For topics related to Machine Learning	An Introduction to Statistical Learning Authors: Gareth James and others (Available online)	

What is Data Science?

- Data science is a **broad field** that encompasses the overall process of **extracting insights** and **knowledge** from data. It involves **collecting, cleaning, organizing, analyzing, and interpreting data** to uncover **patterns, trends, and meaningful information**.
- Data science utilizes various techniques, methodologies, and tools to **extract valuable insights** from data.

Case Study

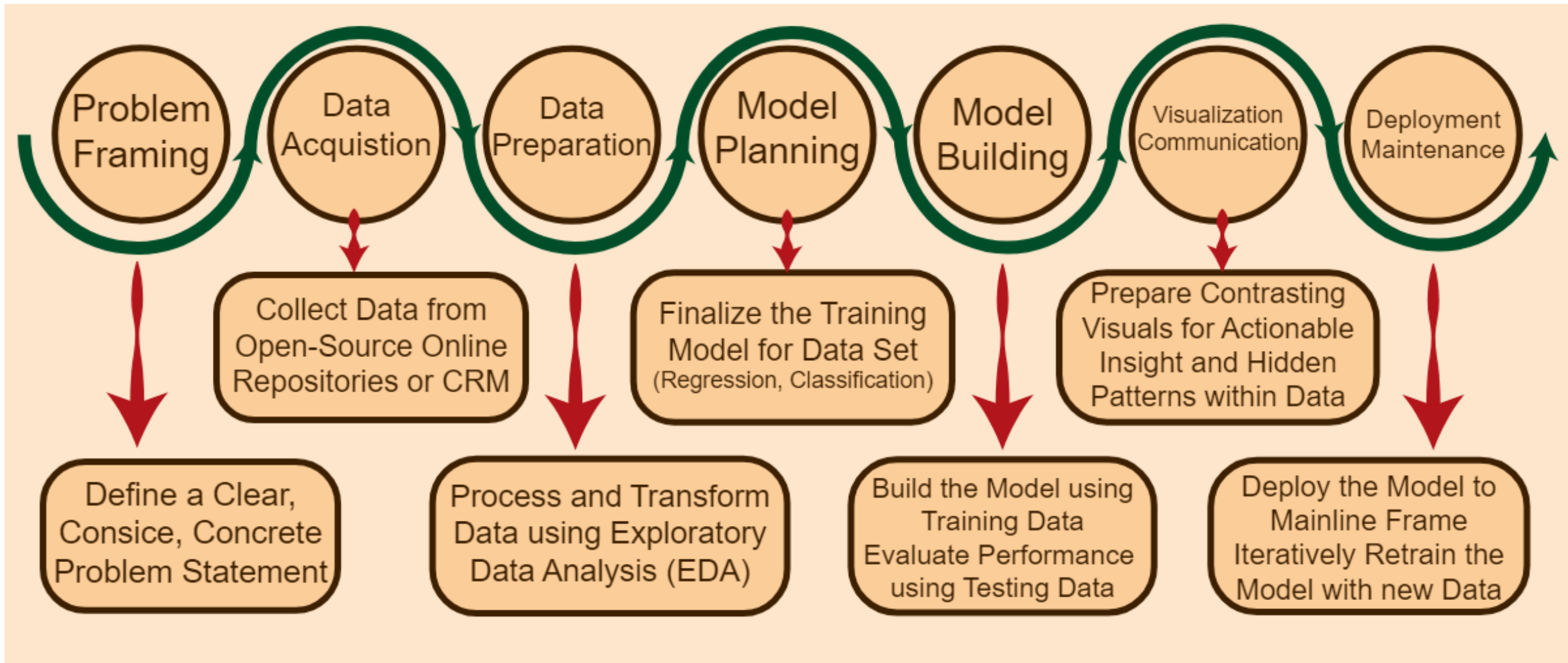
Case Study Overview

Attribute	Details
The problem	Optimization of process(es) in a chemical plant
Goals	1. Operating parameters' optimization 2. Throughput prediction 3. Breakdown prediction 4. Predictive Maintenance of equipment
Basis	Multiple year, daily averages of operational parameters available from Data Acquisition System
Columns	132
Rows	2657 (approximately 7 years' daily averages data)

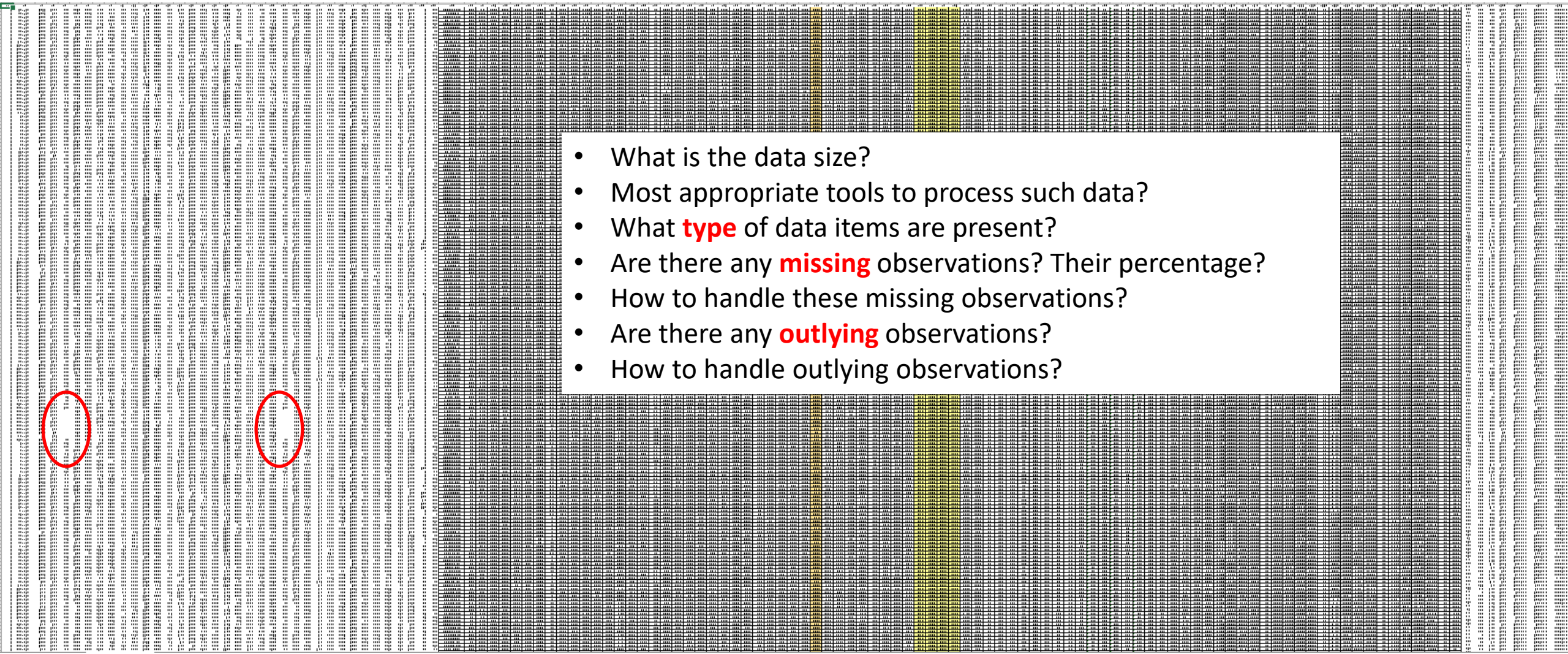
The Data ...

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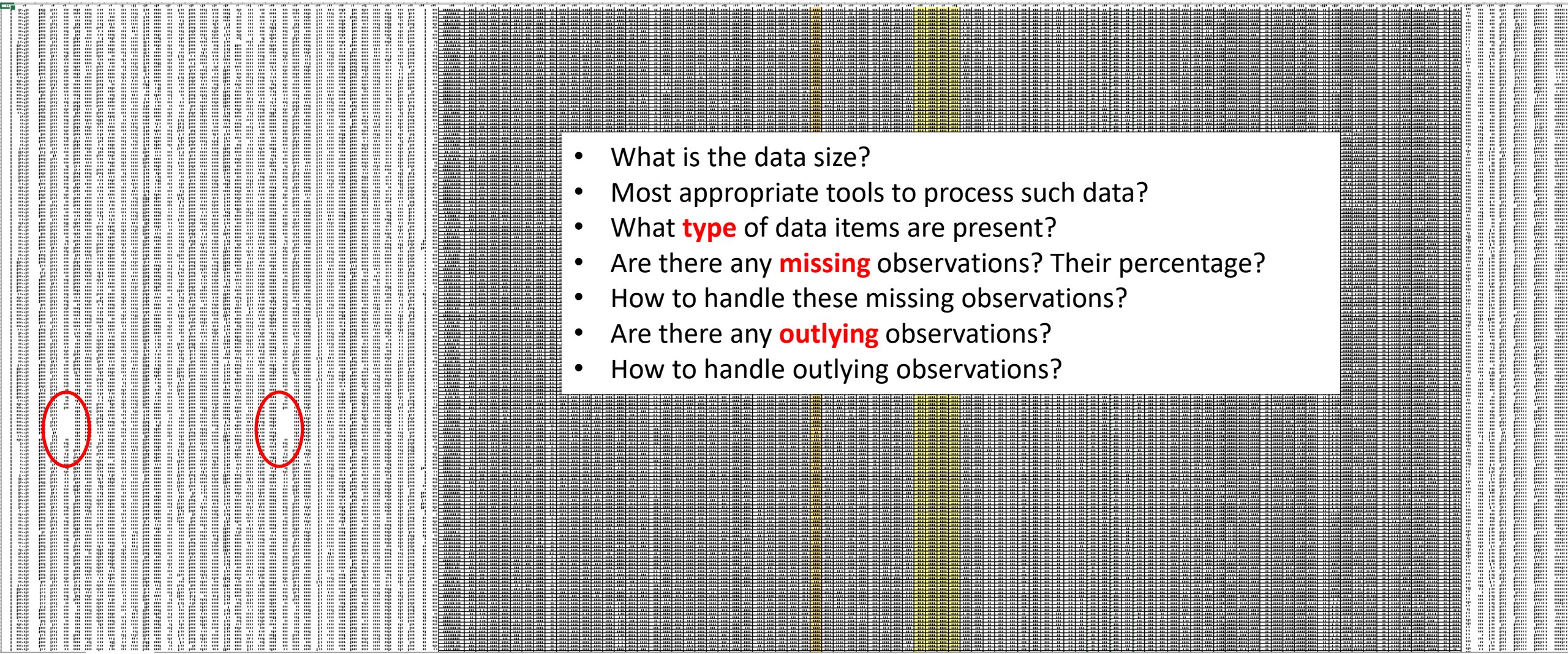
The Data Science Process



Preliminary analysis of the Data



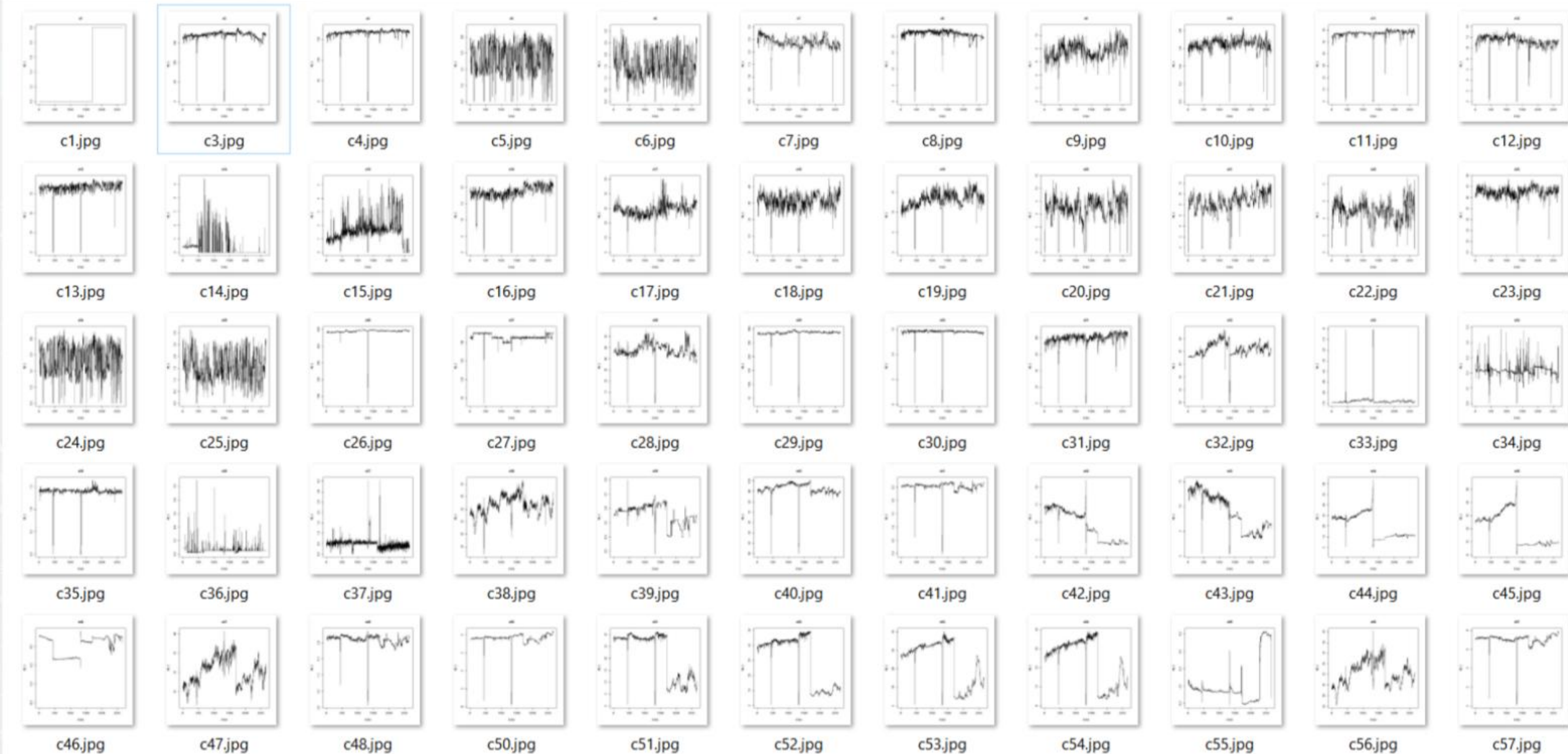
- What is the data size?
- Most appropriate tools to process such data?
- What **type** of data items are present?
- Are there any **missing** observations? Their percentage?
- How to handle these missing observations?
- Are there any **outlying** observations?
- How to handle outlying observations?

- # Preliminary analysis of the Data
- 
- What is the data size?
 - Most appropriate tools to process such data?
 - What **type** of data items are present?
 - Are there any **missing** observations? Their percentage?
 - How to handle these missing observations?
 - Are there any **outlying** observations?
 - How to handle outlying observations?

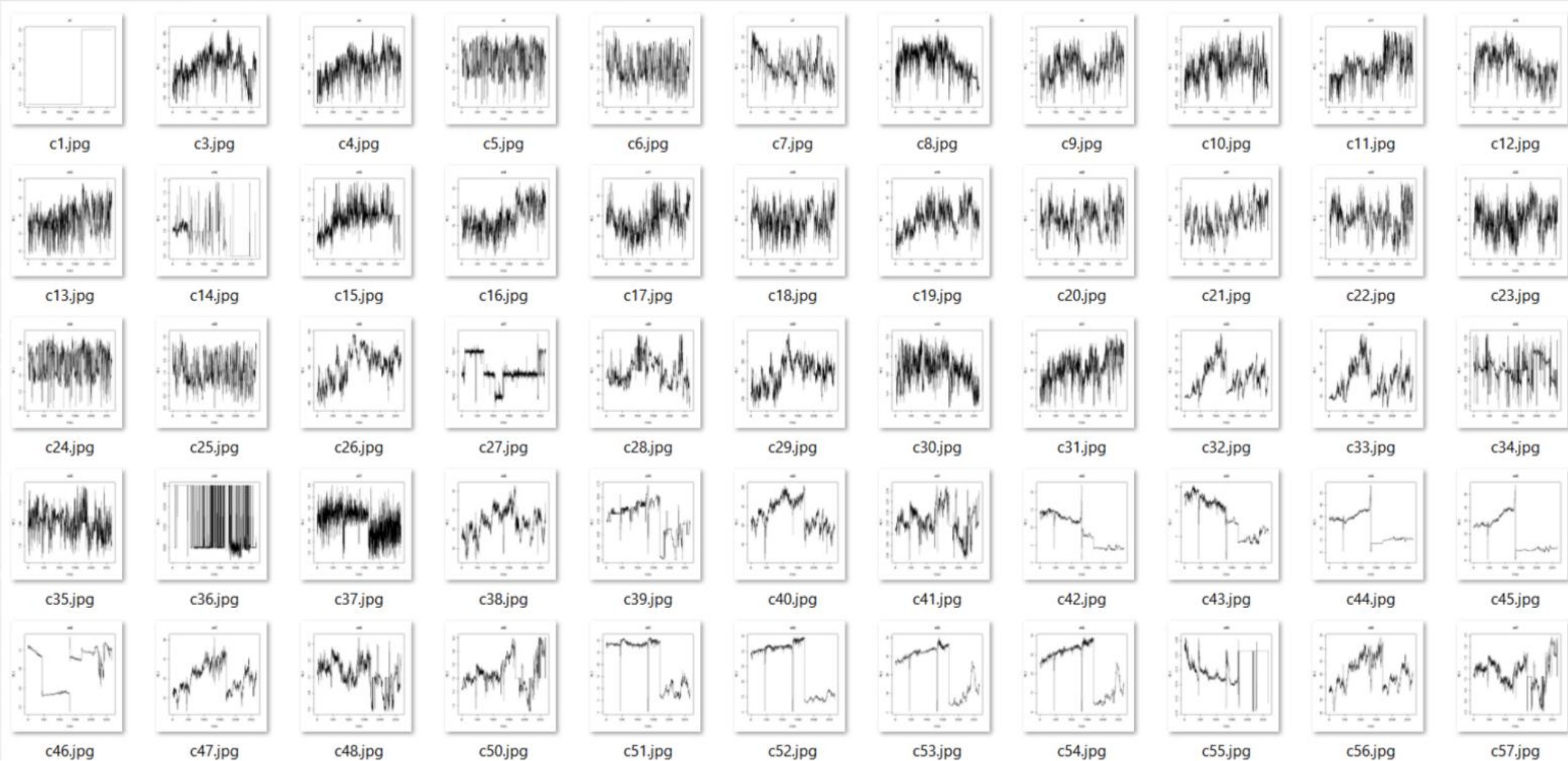
Early analysis / decisions based on Data

- What is the **type** of the problem to be solved?
- Which columns constitute dependent data? The '**Y**' data
- Which columns constitute independent data? The '**X**' data
- Are there any clear dependencies / **correlations** among '**X**'
- Is data scaling / **normalization** required prior to modelling?
- Do some of the columns require **encoding**?
- What is the **modality** / **distribution** of each variable?
- How much of the data should be used for **training** the model
- Can accurate analysis be done by **data sampling**?

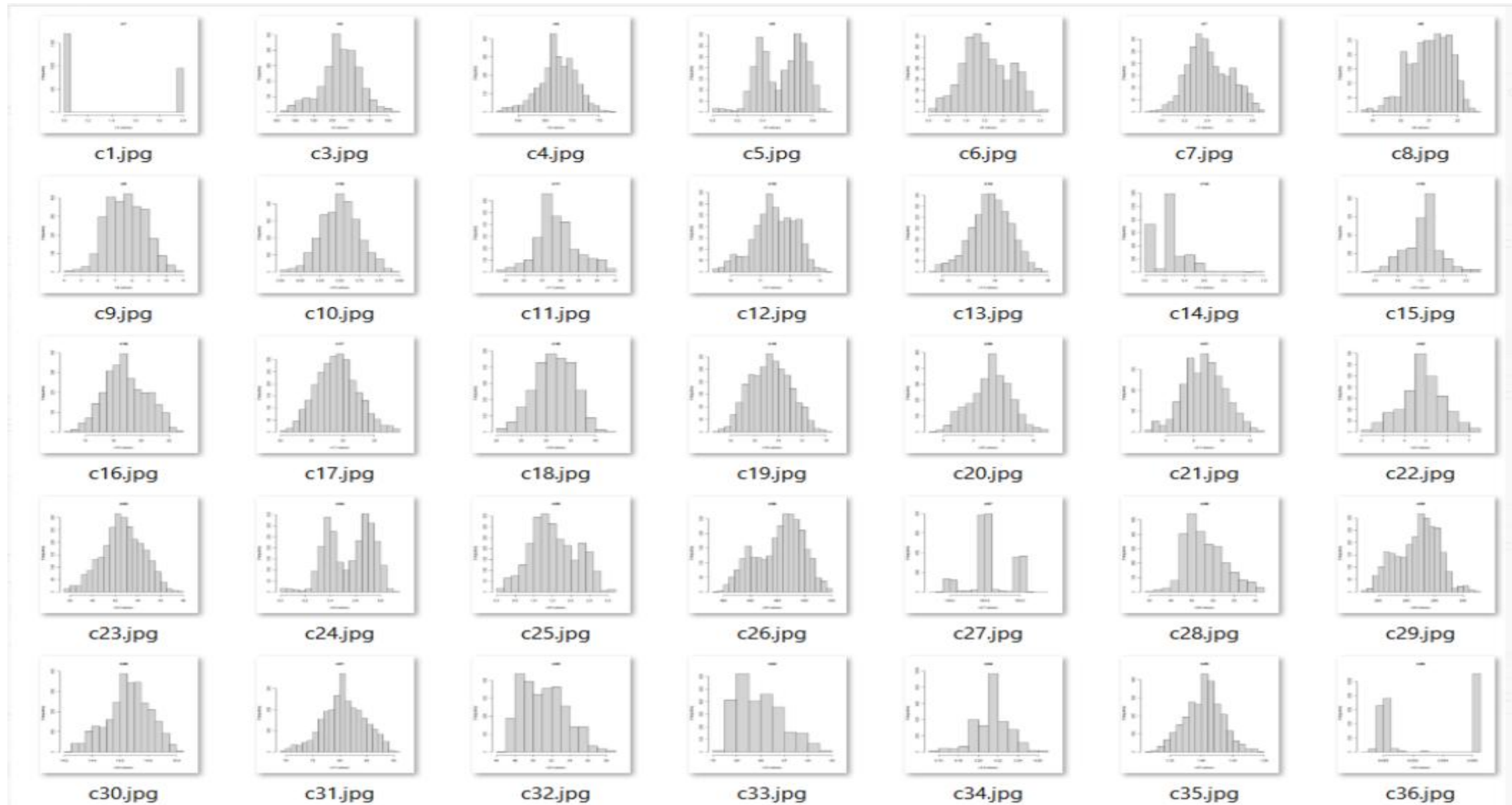
Data plots to understand trends / detect outliers



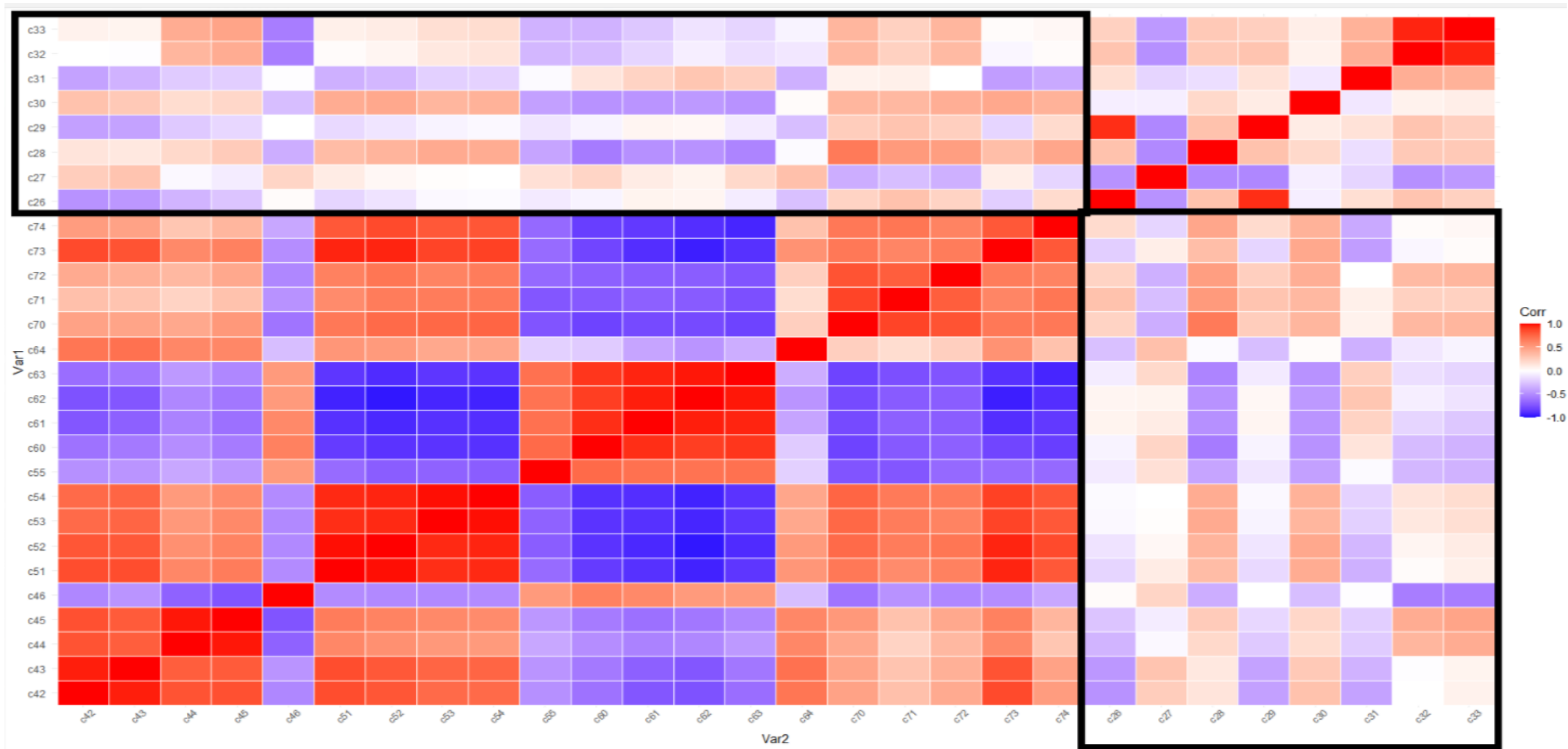
Data plots after **processing** outliers



Modality and Distributions



Correlation between variables



Use of Linear Regression to model and predict a parameter

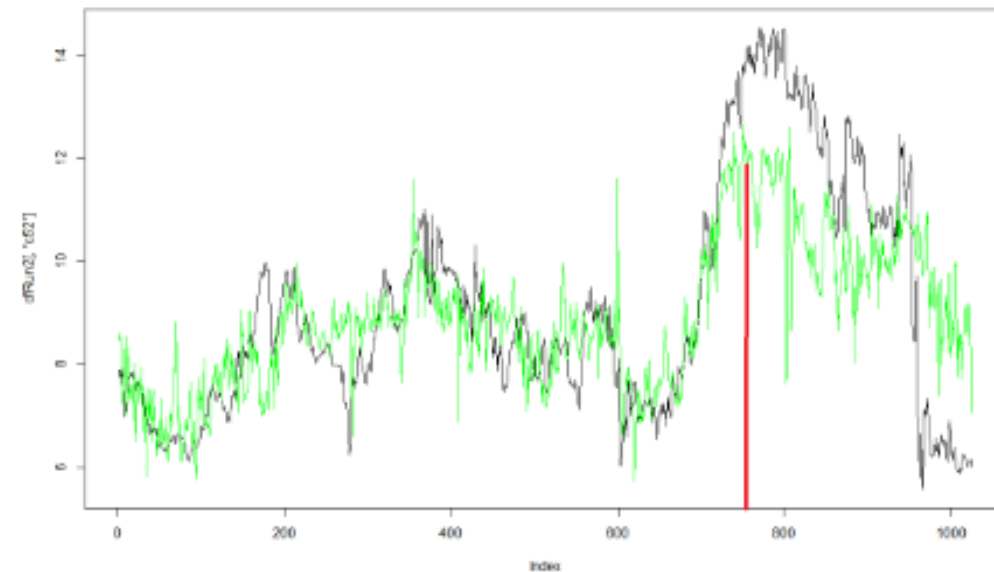
Training: First 750 data points

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-16.209642	5.592980	-2.898	0.003866	**
c161	0.011909	0.001154	10.317	< 2e-16	***
c28	0.192089	0.031109	6.175	1.10e-09	***
c7	1.762212	0.244322	7.213	1.38e-12	***
c17	-0.099812	0.019082	-5.231	2.21e-07	***
c158	0.123261	0.018093	6.813	2.01e-11	***
c160	0.004787	0.001771	2.703	0.007041	**
c39	6.898476	1.100166	6.270	6.18e-10	***
c22	-0.104352	0.034339	-3.039	0.002460	**
c11	-0.122940	0.036342	-3.383	0.000756	***
c15	-0.371219	0.056582	-6.561	1.02e-10	***
c30	1.702216	0.337681	5.041	5.85e-07	***
c23	-0.274084	0.039142	-7.002	5.74e-12	***
c35	5.033540	1.451099	3.469	0.000554	***
c16	-0.381731	0.073873	-5.167	3.07e-07	***
c139	-0.221037	0.036769	-6.011	2.91e-09	***
c31	0.156248	0.022446	6.961	7.55e-12	***
c143	-0.236297	0.034607	-6.828	1.82e-11	***
c157	0.160972	0.038140	4.221	2.75e-05	***
c163	0.009764	0.002779	3.513	0.000471	***
c9	-0.267089	0.070883	-3.768	0.000178	***
c8	-0.369612	0.122558	-3.016	0.002652	**
c10	3.208038	1.541097	2.082	0.037723	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8832 on 727 degrees of freedom
Multiple R-squared: 0.6309, Adjusted R-squared: 0.6198
F-statistic: 56.49 on 22 and 727 DF, p-value: < 2.2e-16



Interpretation of results

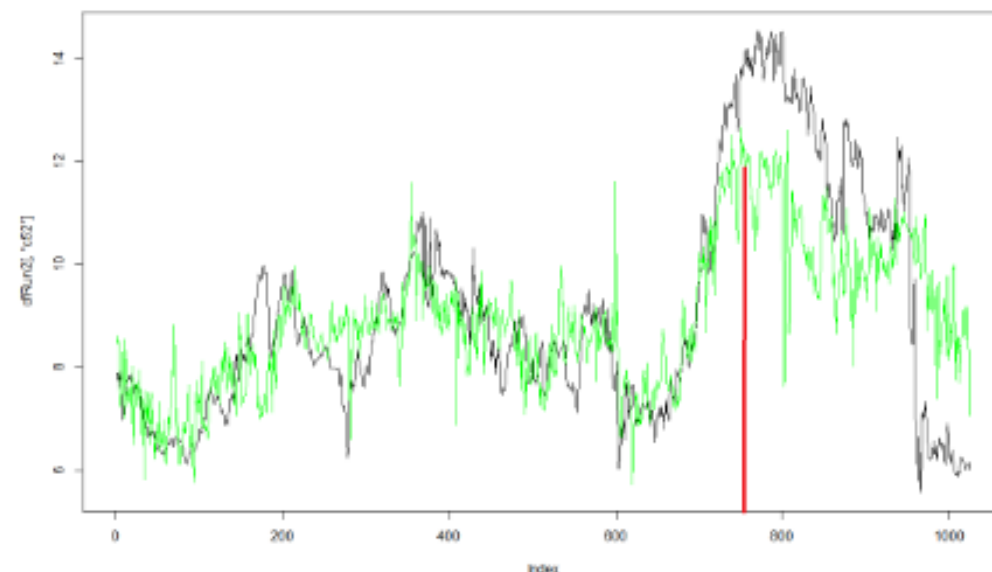
Training: First 750 data points

Coefficients:

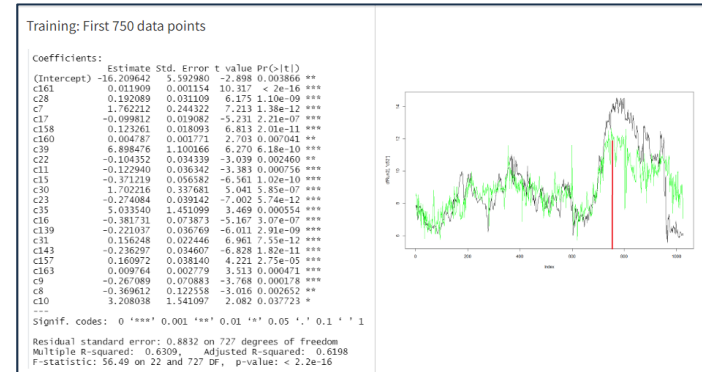
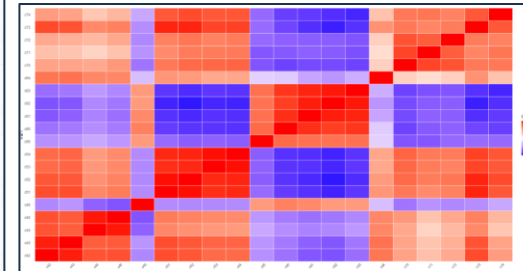
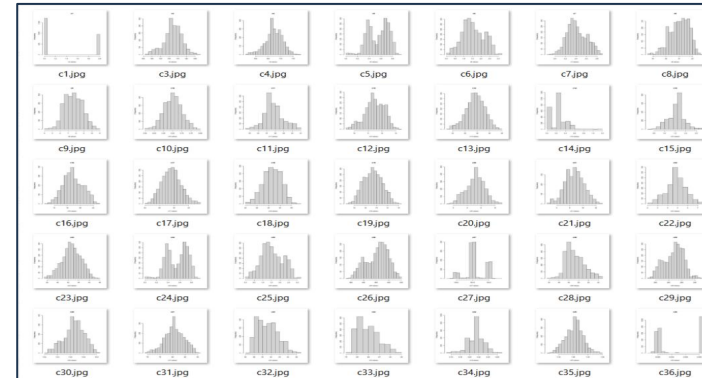
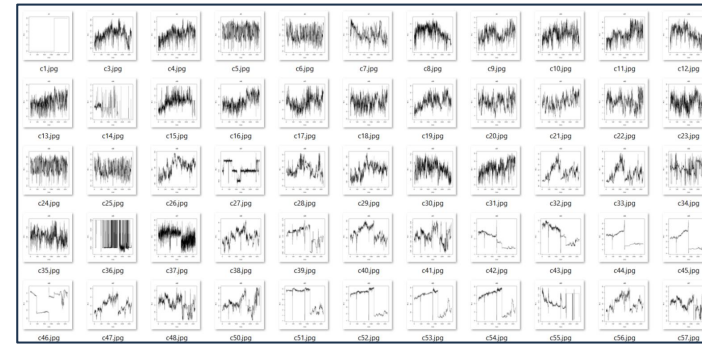
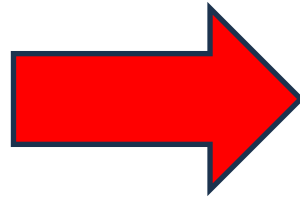
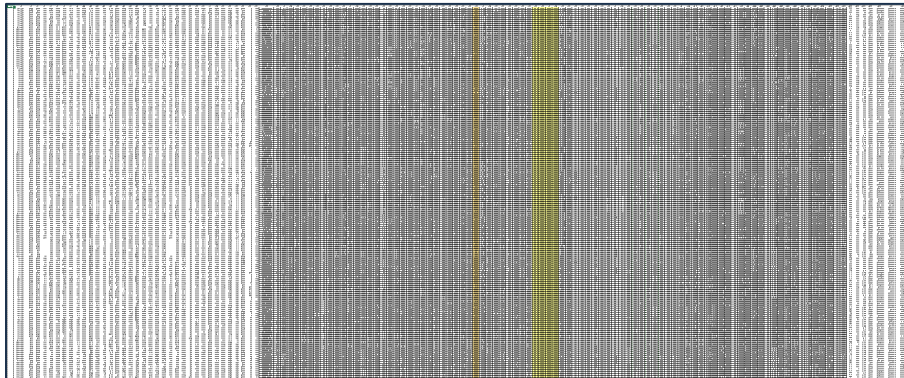
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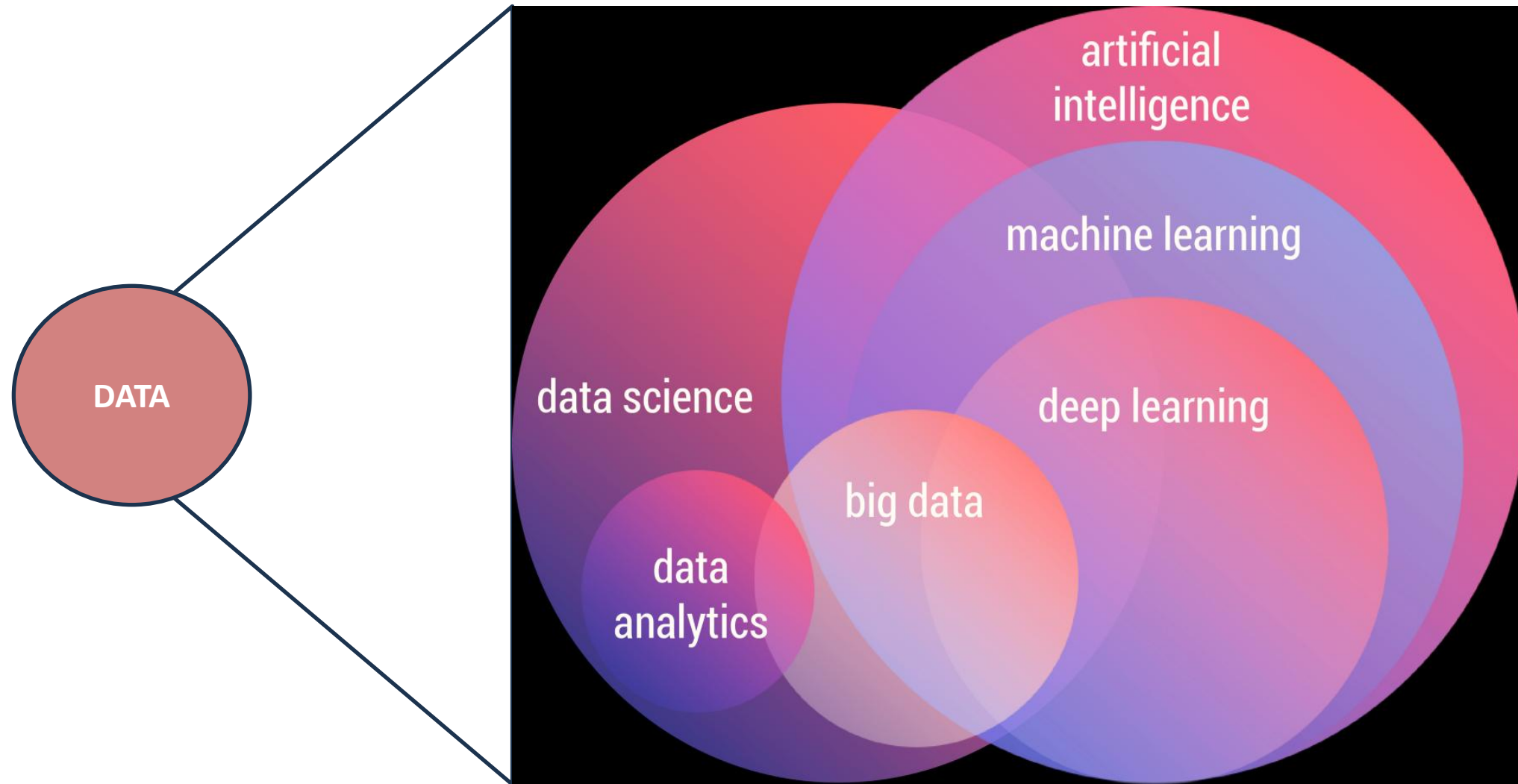
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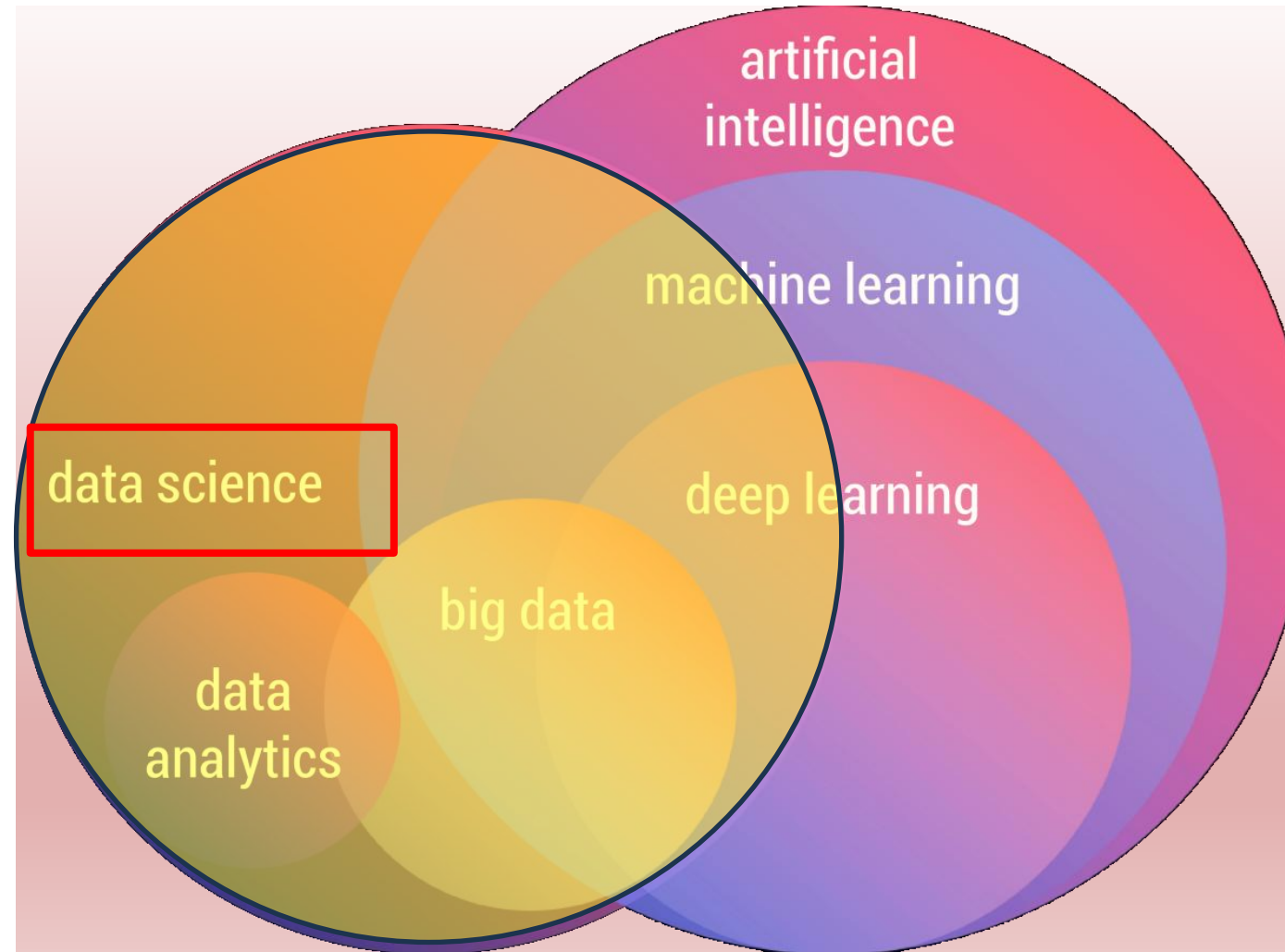


Goal of Data Science : Extract Insights from Data

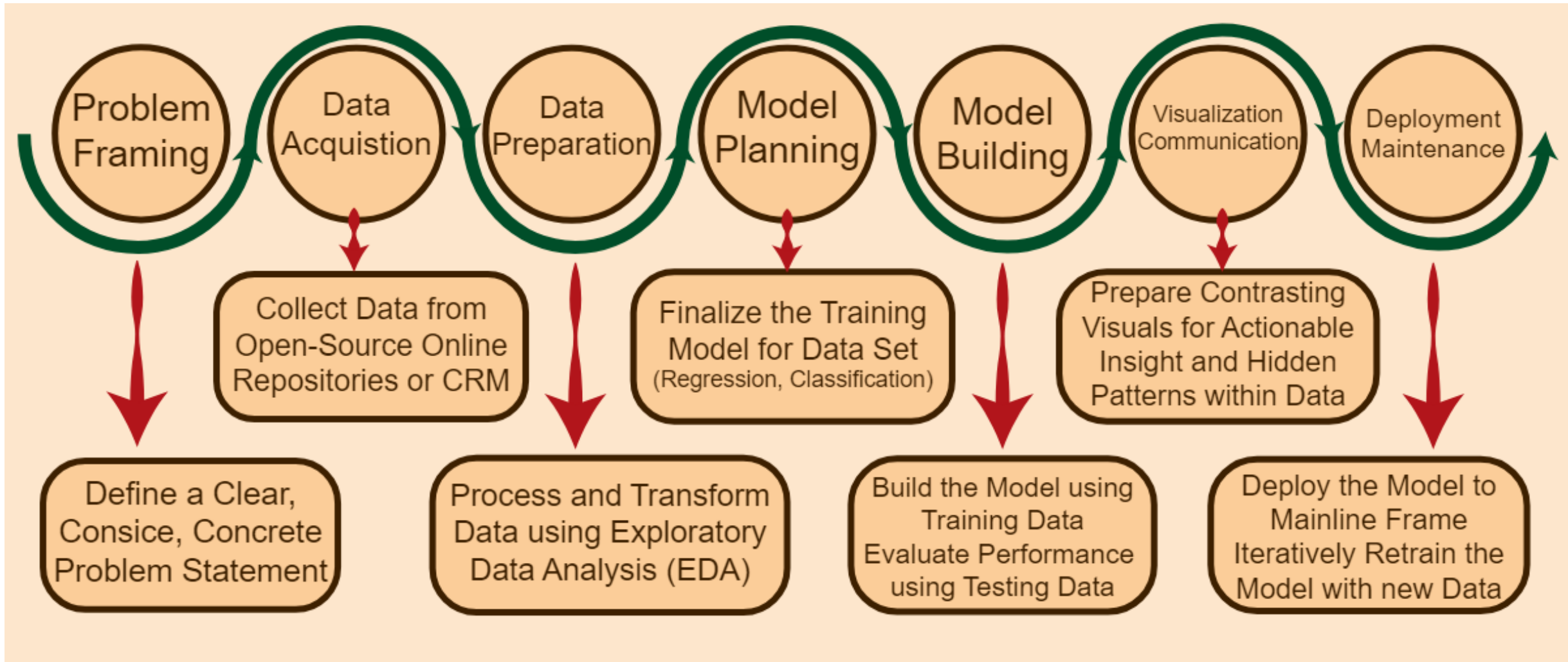


“Where there is Data, There is a way ...”

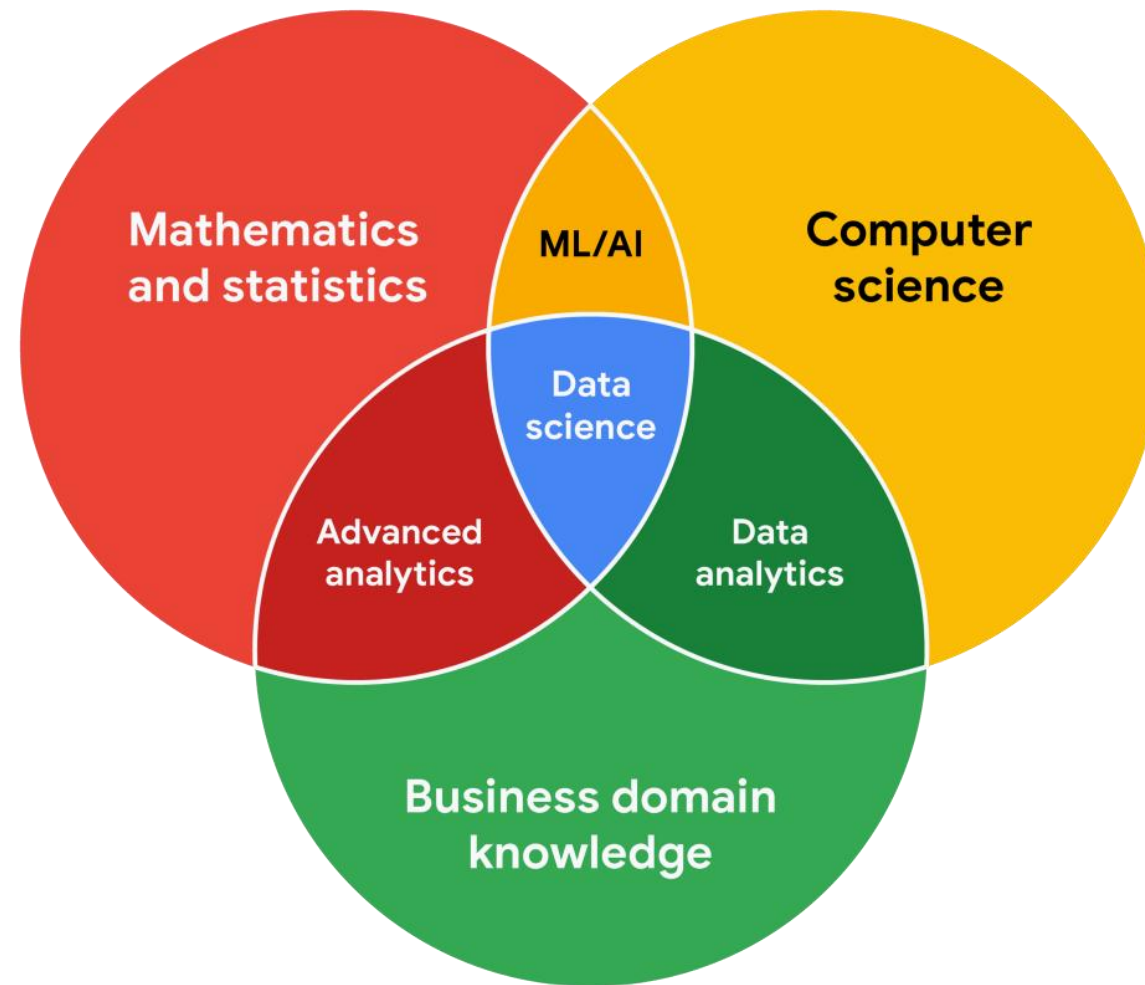




The Data Science Process



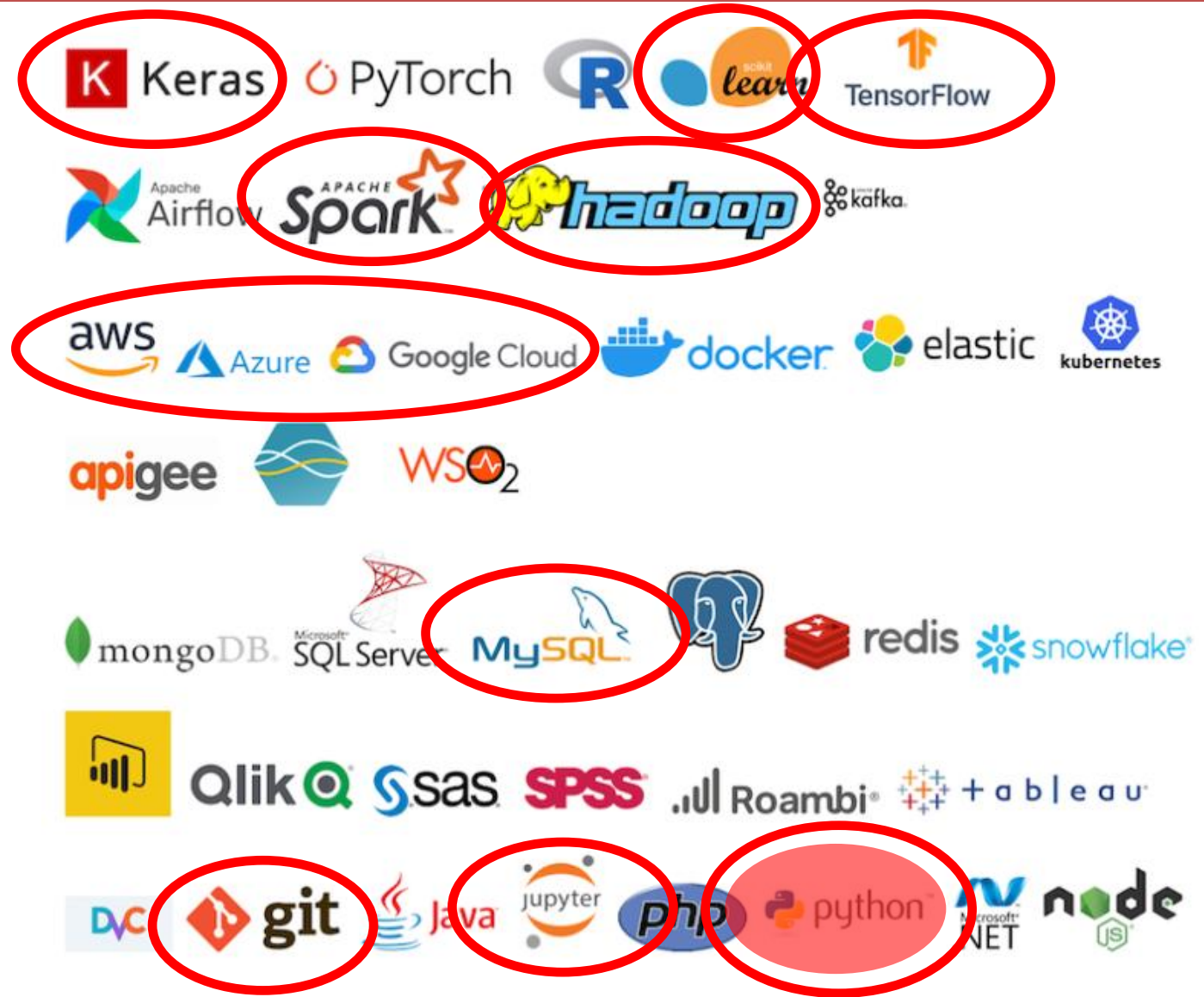
Putting it all together !



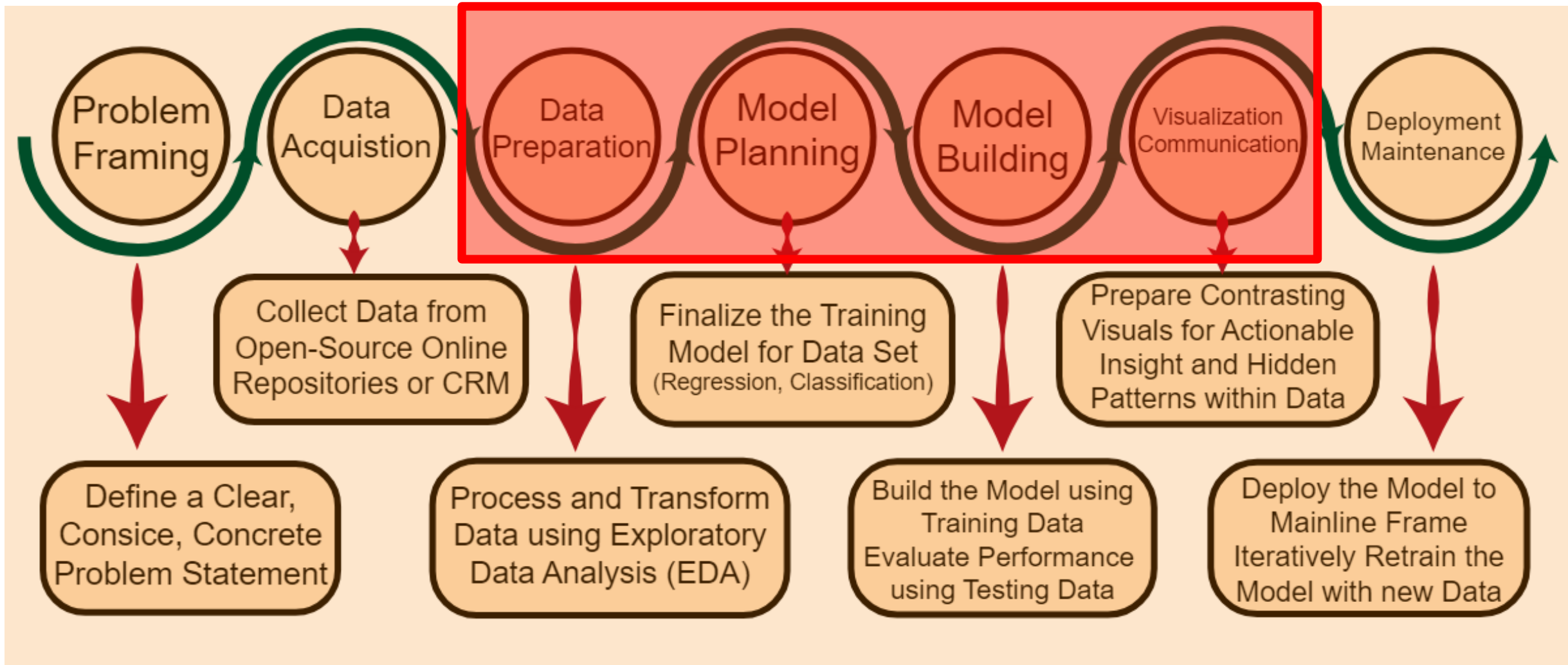
- The nature of ‘programming’ has changed over generations
 - Machine, Assembly, C like, SQL like, AI powered
- Programming, conventionally
 - It is all about syntax, semantics, structure, optimization, etc.
- What does ‘programming’ mean, in the context of Data Science?
 - Understanding and effective use of the libraries / modules / packages
 - Understanding and correct use of various tools
 - Lately: Generating correct problem descriptions for LLMs
 - Prompt engineering

Plethora of tools ... (by no means complete!)

- Machine Learning
- Computer Vision
- Edge AI for Smart Devices and Machinery
- Predictive Analytics
- Data Visualization
- Big Data Engineering
- Open API
- Cloud Technologies
- Robotic Process Automation
- Sentiment Analysis
- Natural Language Processing
- DataOps, MLOps, and DevOps
- Blockchain and Decentralized Ledger Technologies



Availability of tools



Plethora of tools ... (by no means complete!)

- Machine Learning
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